UNIVERSITY CEU - SAN PABLO

POLYTECHNIC SCHOOL

BIOMEDICAL ENGINEERING DEGREE

BACHELOR THESIS

Detection of Mild Cognitive Impairment using Graph Neural Networks and Magnetic Resonance Imaging for Early Diagnosis of Alzheimer's Dementia

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September 2024

UNIVERSIDAD SAN PABLO-CEU
ESCUELA POLITÉCNICA SUPERIOR
División de Ingeniería

Datos del alumno

NOMBRE:

Datos del Trabajo

TÍTULO DEL PROYECTO:

Tribunal calificador

PRESIDENTE:	FDO.:
SECRETARIO:	FDO.:
VOCAL:	FDO.:

Reunido este tribunal el ______/______/______, acuerda otorgar al Trabajo Fin de Grado presentado por Don________________________________la calificación de _________________.

ABSTRACT

Alzheimer’s disease (AD) is the most common and prevalent neurodegenerative disorder today, affecting along with other forms of dementia over 55 million people worldwide. Considering that no effective cure has been discovered yet, a timely diagnosis remains crucial to manage symptoms and slow the progression of the disease. One of the most extensively studied and promising neuroimaging modalities for early AD detection is magnetic resonance imaging (MRI), although its analytical complexity has led to the development of deep-learning methods that can assist neurologists during their observation. The presented project addresses these challenges by proposing a graph neural network (GNN)-based model for the detection of mild cognitive impairment (MCI), a potential precursor of AD dementia. Both structural and resting-state functional MRI data from 152 subjects of the Alzheimer's Disease Neuroimaging Initiative (ADNI) were employed, including 76 cognitively impaired and 76 cognitively normal (CN) individuals. Through this cross-sectional multimodal study, a brain and subject network was constructed using the Neo4j graph database, aimed at identifying cortical thinning and grey matter loss in several interconnected brain regions (ROIs) associated with MCI. Limitations in message propagation, likely due to integration issues, required using a tabular format and led to suboptimal performance.

RESUMEN

La enfermedad de Alzheimer (EA) afecta con otras demencias a más de 55 millones de personas en todo el mundo, siendo el trastorno neurodegenerativo más común y prevalente. Dado que actualmente no existe una cura efectiva, la detección temprana de la EA es crucial para manejar sus síntomas y ralentizar su avance. Una de las técnicas de diagnóstico más prometedoras es la resonancia magnética (RM), aunque su complejidad ha impulsado el desarrollo de métodos de aprendizaje profundo que puedan ayudar a su observación. El presente proyecto aborda estos desafíos proponiendo un modelo basado en redes convolucionales de grafos (GCN) para la detección del deterioro cognitivo leve (DCL), un precursor potencial de la demencia por EA. Utilizando datos de RM estructural y funcional de 152 individuos (76 con DCL y 76 cognitivamente normales) de la iniciativa ADNI, se construyó una red cerebral y de sujetos en la base de datos de grafos Neo4j. El objetivo de este estudio multimodal y transversal fue identificar pérdida cortical y de materia gris en regiones interconectadas del cerebro asociadas con el DCL. Limitaciones en la propagación de mensajes durante el entrenamiento, probablemente por problemas de integración, requirieron el uso de un formato tabular y llevaron a un rendimiento subóptimo.

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1 INTRODUCTION

Dementia stands currently as a public health priority, ranking as the seventh leading cause of death globally [1]. This umbrella term encompasses a wide range of diseases known to alter the cognitive function (memory loss, mobility problems, communication impairment, behavioural issues), eventually interfering with one's social relations and ability to perform daily activities and basic bodily functions, such as walking and swallowing [2,3]. Its most common form is Alzheimer’s disease (AD), contributing to 60–80% of the cases [3].

AD is an irreversible and progressive brain disorder which is characterised by the degeneration of neurons and the accumulation of abnormal proteins, namely beta-amyloid (Aβ) plaques and tau tangles. Unfortunately, AD is generally considered ultimately fatal, with research showing that patients aged 65 and above survive an average of four to eight years after a diagnosis of Alzheimer’s dementia [3]. Despite having identified several risk factors for AD, such as age or presence of the APOE-e4 gene, the exact cause remains unclear. Several drugs have been developed to address the underlying physiopathology of this neurodegenerative disease (aducanumab and, recently, lecanemab [4]), but none has been found to stop its progression. Other treatment approaches, including both pharmacologic and nonpharmacologic strategies (e.g., promoting a healthy diet, encouraging physical activity) aim to improve symptoms and quality of life by tackling potentially modifiable risk factors. With the realisation that very limited treatment options are available for AD while urgent action must be taken to reduce its burden, there has been a shift in focus towards identifying individuals much earlier in the disease process and implementing measures to reduce or prevent further progression [5].

According to the 2018 National Institute on Aging and Alzheimer’s Association (NIA-AA) criteria [6], the Alzheimer’s disease continuum is characterized by three general stages: cognitively normal or unimpaired (CN), mild cognitive impairment (MCI, also known as prodromal AD), and dementia (see Figure 1). The first stage corresponds to a preclinical, asymptomatic phase where measures of pathological Aβ and tau biomarkers are observed on cerebrospinal fluid (CSF) or positron emission tomography (PET) [1]. This can last for 20 to 30 years, with some patients remaining in a cognitively normal stage for the rest of their lives, but others eventually manifest measurable symptoms such as problems with memory, language and thinking that meet the criteria for MCI. According to a 20-year retrospective study on memory clinic patients, not all individuals with MCI due to AD will inevitably progress to dementia, but approximately 32% of those diagnosed do, with a median time to progression of 2 years [7]. Although both pre-dementia stages offer very valuable insight into the nature of AD for the study of preventive measures, MCI has become the biggest target [8]: long-term pharmacological treatment of CN individuals has revealed potentially significant and serious side effects (e.g., amyloid-related imaging abnormalities), making it important to identify and treat only subjects having the highest risk of developing clinical AD within a short time frame [9].

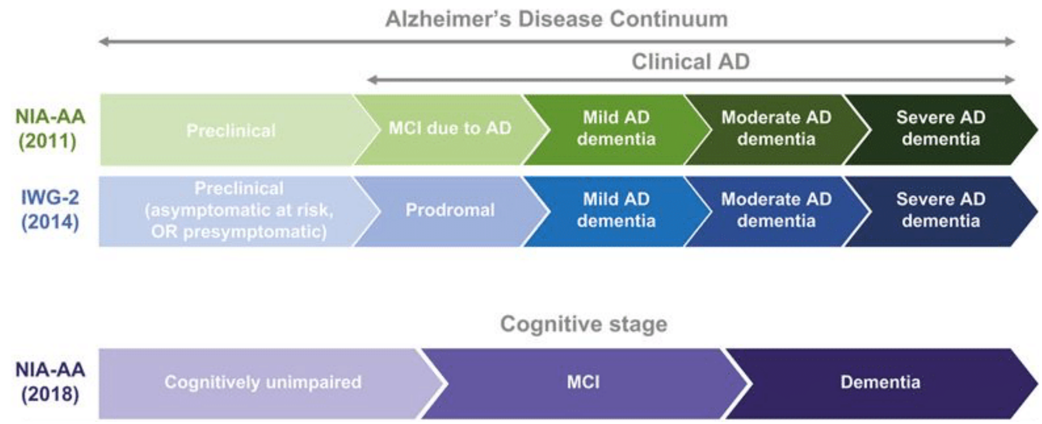
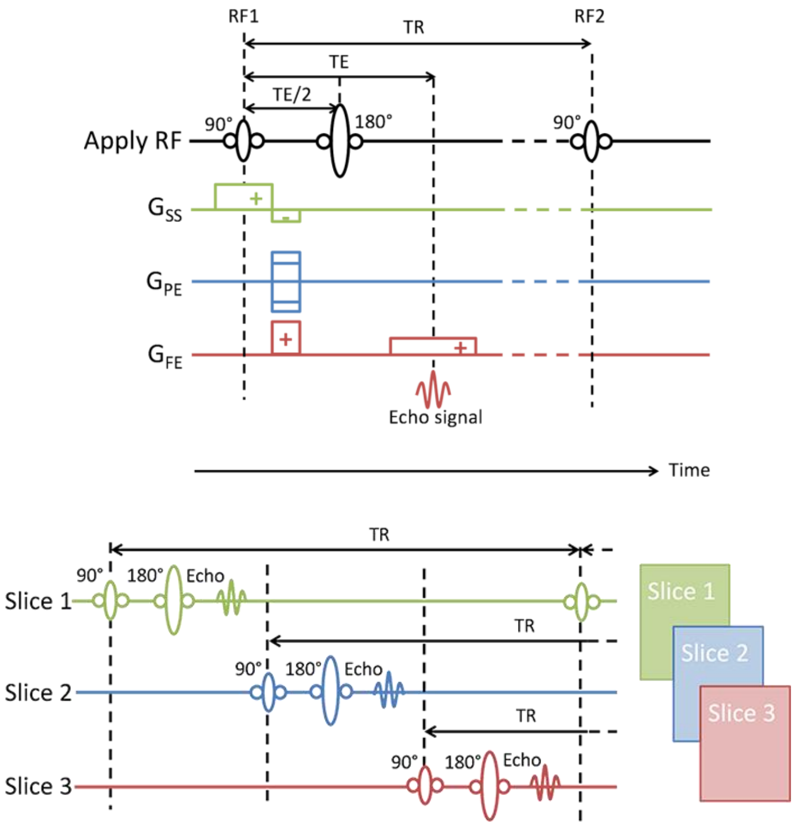


Figure 1. Comparison between AD disease continua as defined by NIA-AA 2011, IWG 2014, and NIA-AA 2018, which serves as the framework for this paper [10].

Whereas the recognition of the AD neuropathology is highly standardized through CSF biomarker assays and PET scans, different diagnostic criteria have been proposed and modified over time for MCI, specifically [11]. Latest guidelines from the NIA-AA for diagnosis of MCI due to AD recommend combining clinical evaluations of cognitive impairment with biomarker analysis for Aβ deposition and neuronal injury or degeneration [12]. Neurodegeneration includes increased levels of CSF tau/ phosphorylated-tau, and several structural and functional measures obtained through diagnostic imaging methods, such as structural magnetic resonance imaging (MRI), fluorodeoxyglucose (FDG)-PET scanning, diffusion tensor imaging (DTI), or functional

MRI (fMRI). Topographical analysis through sMRI has been particularly employed in this research area due to its non-invasion, high resolution, fine tissue contrast, and moderate cost, despite data being less available [12].

The basic principle of all MRI modalities relies on the property of hydrogen nuclei to absorb radio frequency (RF) energy when placed in a strong magnetic field [13]. After first aligning these protons in the brain through static magnets, RF pulses are applied to manipulate their magnetization levels, initially with a dephasing 90º pulse and later with a rephasing 180º pulse. As their spins return to equilibrium during the so-called T_1 and T_2 relaxation, a spin echo is generated, whose peak signal occurs at a specific point known as the echo time (TE). This emitted energy is then detected by a receiver coil, while several gradients are used to locate the nuclei spatially, and the pulse cycle or "spin-echo sequence" is repeated for all brain tissue slices according to the repetition time (TR or time between successive 90º RF pulses) (see Figures 2 and 3).



Figures 2 and 3. Signal acquisition for a single brain slice (above) and spin-echo sequence for three brain slices (below). G_{SS}, G_{PE} and G_{FE} denote the three types of gradients responsible for slice selection, phase encoding and frequency encoding, respectively. [13]

By adjusting the TE and TR parameters and observing differences in T_1 and T_2 relaxation times between tissues (fat, muscle), image contrast is created. Together with the spatial encoding process, this allows for the three-dimensional reconstruction of detailed brain structures, which can help identify several volumetric and surface biomarkers associated with MCI. These include grey matter decay [14] (see Figure 4) and atrophy of the hippocampus and association cortex: lateral temporal, posterior cingulate, inferior parietal, praecuneus, medial frontal gyrus and frontal cortex [15,16]. The shrinkage of these cortical regions (also, “cortical thinning”) is evaluated as a measure of their cortical thickness (CT), usually calculated as the distance between white matter and pial surfaces [17].

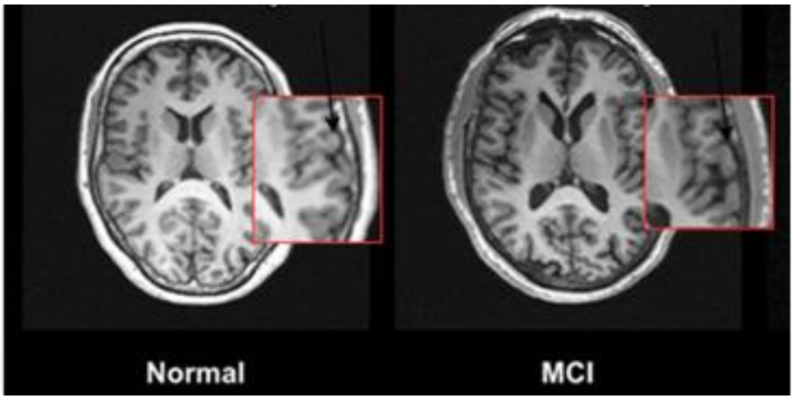


Figure 4. MRI showing grey matter volume (GMV) of a healthy control and an MCI patient [14]

Additionally, recent studies show that functional connectivity (FC) derived from resting-state functional MRI (rs-fMRI) data has become a powerful approach to understand the pathology of brain diseases [18]. The physical principle behind rs-fMRI is essentially the same as for sMRI, but it measures magnetization changes between oxygen-rich and oxygen-poor blood flow (namely, low-frequency blood-oxygen-level dependent or BOLD signals) to detect active and inactive brain areas instead [19]. With a large focus on the default-mode network (DMN) including the posterior cingulate cortex, praecuneus, and medial temporal lobes, studies have found deficient network connectivity in patients with cognitive dysfunction compared to normal controls [20].

However, MRI presents challenges, including the need for specialized expertise and high measurement variability in sMRI across individuals, which hinders its effectiveness as a stand-alone for early AD detection [21]. Similarly, fMRI encounters issues with low sensitivity and specificity, and difficulties in longitudinal studies of progressive dementias, especially in cases of severe cognitive impairment because of high head motion [20, 21]. To address these limitations, multimodal neuroimaging studies have emerged as a promising alternative. By combining quantification of clinical features and brain activation patterns, these hybrid approaches demonstrate general superior diagnostic performance of AD/MCI compared to unimodal studies, despite smaller sample sizes and limited generalizability to diverse populations [22].

Another acknowledged problem with these neuroimaging modalities is that MRI images are complex and often noisy, making it very difficult for the human eye to discriminate specific features of MCI from those of other neurological disorders or AD stages [23]. To address this issue, several deep learning algorithms have been proposed in recent years to enhance image clarity and improve feature extraction.

Deep learning is a type of machine learning technique that aims to imitate human reasoning in data processing and pattern recognition tasks to solve complex decision-making problems [23]. Unlike traditional machine learning, these models can automatically learn relevant features from raw data without the need for explicit feature engineering, which can be time-consuming and requires expert knowledge [14]. Even when feature extraction is performed, their architecture provides the ability to generalize and learn complex relationships within the data. Together with the rapid increase in high-volume datasets, the advances in these artificial intelligence methods have revolutionised performance in numerous areas, with convolutional neural networks (CNNs) being the most successful kind regarding organ segmentation and disease classification in neuroimaging [23].

But despite their strong capability to extract spatial and temporal features, CNNs are inherently designed to operate on Euclidean data, which makes them less suited for modelling irregular and more arbitrary geometric structures, such as brain connectivity [24]. Furthermore, the relationships between data points in multimodal data often cannot be adequately captured using linear methods, which is a particularly relevant limitation for CNNs due to their reliance on grid-like structures and local connectivity patterns [20].

Consequently, graph methods have emerged to tackle these issues by representing each equally-sized voxel or region of interest (ROI) in the brain by a node, and the information between two nodes, by an edge. Graph neural networks (GNNs) are one of the deep learning techniques that integrate this graph architecture into their design, and by means of spatial convolution, they aggregate and transform the features of adjacent nodes to extract information on their relations [25] (see Figures 5 and 6).

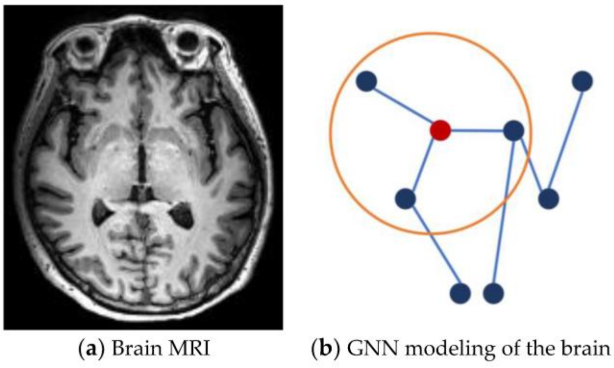


Figure 5. (a) T1-MRI slice of the brain. (b) Schematic diagram of GNN modeling for the brain [25].

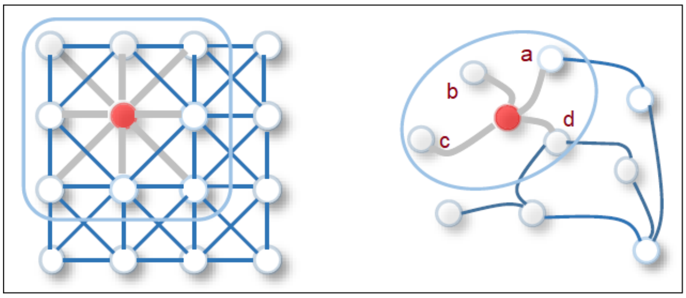


Figure 6. Visual representation of a graph in a Euclidean space and in a non-Euclidean space to differentiate between data representation in CNNs and GNNs, respectively [26].

Unlike voxel-based models, which treat each 3D pixel in the imaging data as an individual node in the graph, GNNs that are developed within a ROI-based approach selectively target brain regions known to be affected by AD pathology [23]. This approach to feature extraction offers a particularly interesting balance between sensitivity and interpretability for those studies that rely on prior hypotheses of disease pathology or brain atlases, like the Automated Anatomical Labelling (AAL) atlas. Moreover, ROI-based methods significantly reduce the computational burden associated with analysing vast amounts of data by decreasing the feature dimensionality.

Therefore, ROI-based approaches are well-suited to the current context of AD research, where critical regions like the hippocampus and other previously mentioned AD-affected areas, are often the primary focus.

But graph methods do not limit their capacities to just these “individual” or “subject graph” types, focusing on the representation of the brain and its functionality. They also expand them to a macro perspective, namely “population graphs”, where each node can represent a single subject, and edges can capture similarities or relationships with other subjects based on imaging or non-imaging features [25]. Some studies go even a step beyond and explore heterogeneous graphs, combining both subject classification and region connectivity. For instance, Y. Gu et. al [27] recently proposed a heterogeneous GNN model with excellent performance in depression classification, utilizing a functional brain connectome graph to identify biomarkers, together with a heterogeneous population graph built from imaging and non-imaging subject data.

In the domain of MCI classification and AD prediction, GNNs have also been investigated under several multimodality approaches, such as the use of sMRI and DTI for brain age estimation, or the application of MRI, PET, and other non-image information of multiple subjects for predicting the conversion of MCI to AD [24, 25]. Graph convolutional networks (GCN) have been particularly utilized as the most successful modality in the study of neurodegenerative diseases [25]. They adapt the convolution operation from CNNs to graphs, aggregating information from a node's immediate neighbours to learn node-level features and ensuring that each neighbouring node contributes equally to the feature representation. For example, J. Liu et. al [28] proposed a GCN framework for Early Mild Cognitive (EMCI) detection using GMV and shortest path length extracted from T1-weighted MRI and rs-fMRI data. The resulting framework (Figure 7) achieved an accuracy of 84.1%, outperforming some existing methods based on support vector machines and dynamic CNNs. Similarly, another GCN-based approach designed by X. Li et al. [29] demonstrated improvements in classification accuracy, sensitivity and specificity for brain disorder identification against several baseline methods. Their Multi-scale Mutual Template-GCN architecture consisted of the integration of multiple levels of spatial information through different brain partitioning templates to generate functional and structural connectivity matrices.

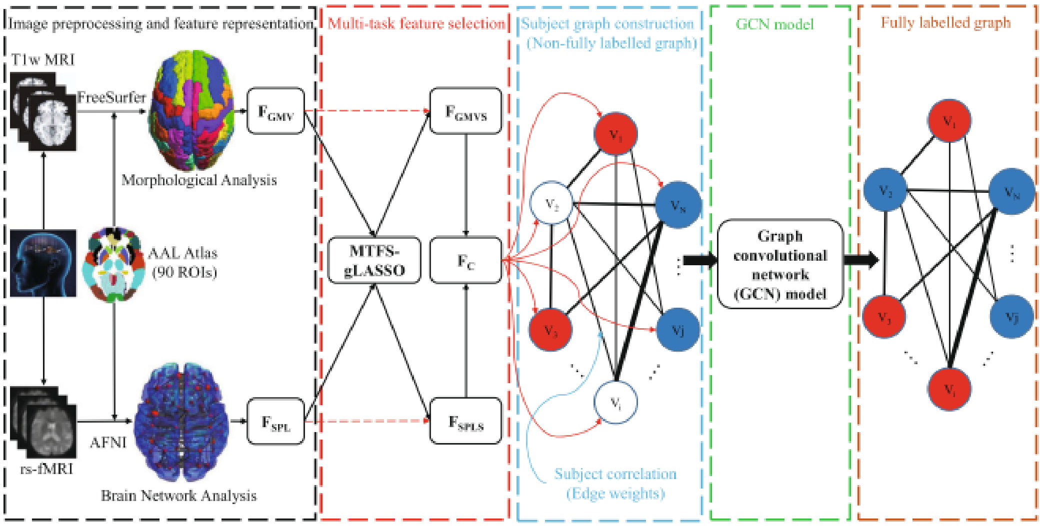


Figure 7. Schema of the multimodal GCN framework for EMCI identification by J. Liu et al. [28], including multi-task feature selection (MTFS)

Over time, the demand for more dynamic and computationally faster models led to the introduction of new GNN-based methods, such as GAT and Graph-SAGE [24]. These inductive algorithms stand out for their ability to adapt to graph geometry changes occurring during model training, adjusting new edge weights or handling new node connections, respectively [25]. Graph-SAGE offers a simpler and more effective approach, having been applied to the study of early AD diagnosis and MCI conversion prediction, even [30].

But integrating complex, multimodal data as graphs into ROI-based GNN models presents a significant challenge, overall. Researchers have explored various approaches to integrate and standardize diverse graph datasets, with Neo4j standing out as one of the most promising graph database management systems, due to its ability to efficiently handle large, heterogeneous data and support sophisticated queries through its graph query language Cypher [31]. Besides, Neo4j employs a property graph model that supports edge attributes, allowing for rich modelling of relationships between entities, and making it especially suitable for neuroscientific research. For instance, in a study by P. Yin et al., Neo4j was used to define standardizing clinical trial eligibility criteria for AD trials [32]. By linking these criteria from different databases with real-world electronic health records and filtering eligible patients through Neo4j queries, they successfully generated cohorts of potential participants. Another significant application of Neo4j in AD research involved creating a comprehensive knowledge graph that extended the Alzheimer’s Disease Ontology for Diagnosis and Preclinical Classification (AD-DPC) [33]. By using Neo4j’s visualization and traversing tools, researchers managed and analysed diverse datasets including diagnoses, psychometric findings, blood and cerebrospinal fluid (CSF) results.

1.1 OBJECTIVES

Building upon the critical role of targeting early dementia diagnosis, the potential of deep learning, and the advantages of multimodal ROI-based methods, this project proposes a novel approach for MCI classification integrating GCNs and Neo4j:

1. A Neo4j graph database was built to efficiently manage and analyse cross-sectional data of both cognitively impaired and healthy samples from the Alzheimer's Disease Neuroimaging Initiative (ADNI). More specifically, a single graph composed of 76 MCI and 76 CN subjects, and 200 regions from the 2018 Schaeffer atlas were constructed using T1-sMRI, rs-fMRI, and other non-image data (sex, age, diagnosis).

2. To connect subject and region nodes in a heterogeneous graph, structural and functional images underwent an extensive preprocessing. Structural images of each subject were first processed through the SPM12 software to extract regional cortical thickness (CT) and grey matter volume (GMV). Then, functional connectivity (FC) between brain regions were obtained through the aggregated pairwise correlations of their BOLD time series across subjects within each group (CN and MCI).

3. These networks were trained and tested through a GNN-based model using the GCN and SAGE algorithms from PyTorch Geometric libraries, with the aim to classify CN and MCI subject nodes.

With the expected results, this project hopes to not only contribute to improving diagnostic strategies for AD, but also serve as a basis for future studies in the field of neuroscience.

2 MATERIAL AND METHODS

An agile methodology was followed during the whole design and development of this project. Its theoretical framework, as well as the specific steps undertaken, will be detailed in this chapter. Access to the repository of this project can be found here: https://github.com/IsabelMartinez2022/DetectionOfMCI_TFG

2.1 DATA ACQUISITION

This research employs a dataset of 70 male and 82 female participants, aged 56 to 96 years old, from the ADNI-3 cohort. Patients in this group were selected based on their initial cognitive degeneration assessments that were conducted between 2017 and 2020, as part of ADNI's longitudinal, multicentre study.

ADNI is an American public/private initiative that was launched in 2003 with the primary goal of testing whether serial MRI, PET, genetic, biospecimen, and clinical and neuropsychological assessments can be combined to measure the progression of MCI and early AD [34]. It is the most referenced dataset among studies that target AD detection through neuroimaging and deep learning, and it was therefore chosen as the data source of this project. ADNI's criteria for MCI includes those participants who "have reported a subjective memory concern either autonomously or via an informant or clinician" but there are no significant levels of impairment in other cognitive areas or any signs of dementia [35]. However, these measures are imperfect "as they are subject to high test-retest variability and are influenced by factors other than changes due to AD" [36]. Another limitation of the ADNI's database is that both the cognitively normal and MCI groups are pathologically heterogeneous regarding AD pathology, so exclusively targeting MCI patients that meet the AD biomarker criteria for our study was not possible.

The sMRI scans in this study consist of T_1-weighted images that were captured at a field strength of 1.5 Tesla in the sagittal plane using three different sequences: Inversion Recovery Fast Spoiled Gradient Recalled (IR-FSPGR), Magnetization Prepared Rapid Acquisition Gradient Echo (MPRAGE) and MPRAGE with reduced distortions (MPRAGE_ND). On the other hand, the rs-fMRI images were obtained during the same session in the axial plane using a 3-T magnetic field, with a resting state protocol where participants were instructed to keep their eyes open and remain still. The following parameters were also specified to ensure robustness: a posterior to anterior (P>>A) phase encoding direction, 3.4 mm of slice thickness, 3000 ms of TR, and 30 ms of TE. Figure 8 shows the raw images of a sample subject as an example:

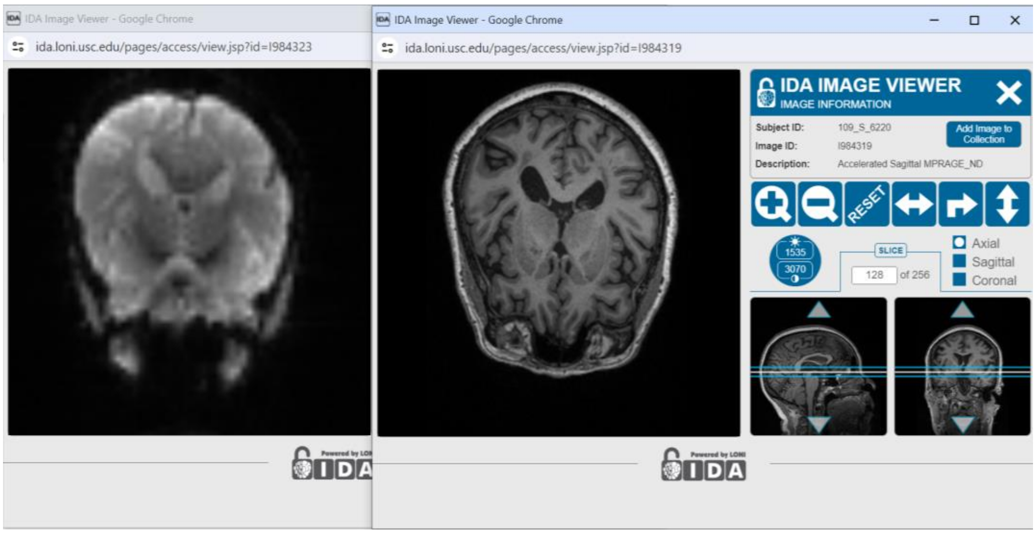


Figure 8. fMRI (left) and sMRI (right) data from subject 109_S_6220 of the ADNI database [34]

2.2 DATA PREPROCESSING

To extract the MCI biomarkers that will define the relations between subjects and regions, a series of image-processing tasks had to be conducted on the acquired data to ensure quality and valid statistical results. Usual techniques for sMRI preprocessing include intensity normalization, registration, skull-stripping, tissue segmentation, and class balancing [37]. On the other hand, the standard preprocessing pipeline for f-MRI consists of realignment and unwarping, slice-timing correction, normalization, outlier identification, and functional smoothing. Although these can be obtained manually, their reliability under test-retest conditions is poor so semi-automated and automated algorithms have become the preferred choice for researchers [17].

One common MATLAB suite that covers techniques for both morphological and brain network analysis is Statistical Parametric Mapping (SPM) [38]. How this free and open software was employed in MATLAB R2023b for the project will be explained in the following sections, differentiating between structural and functional images:

2.2.1 Structural MRI Preprocessing

The SPM package was originally designed for the analysis of functional imaging data from PET, but current versions integrate extensions for computational anatomy to obtain surface measures as well. This is the case of the CAT12 toolbox, which allows for voxel-based morphometry (VBM), surface-based morphometry (SBM), deformation-based morphometry (DBM), and region-based morphometry (RBM) [39]. CAT12 adapts the surface registration techniques from the most popular open-source software for CT estimation, called Free Surfer [40], but is uses a volume-based approach for cortical surface reconstruction. Instead of using the classical measure of distance between the white and pial anatomical surfaces, it relies on projection-based thickness (PBT), which consists of creating CT maps using the information of blurred sulci that focus on the 50% distance boundary ("central surface") between both interfaces (see Figure 9). This method requires less computational effort and allows for a fast and easy-to-use alternative that can compute cortical surface reconstruction of T1-weighted images in any operating system, in about an hour.

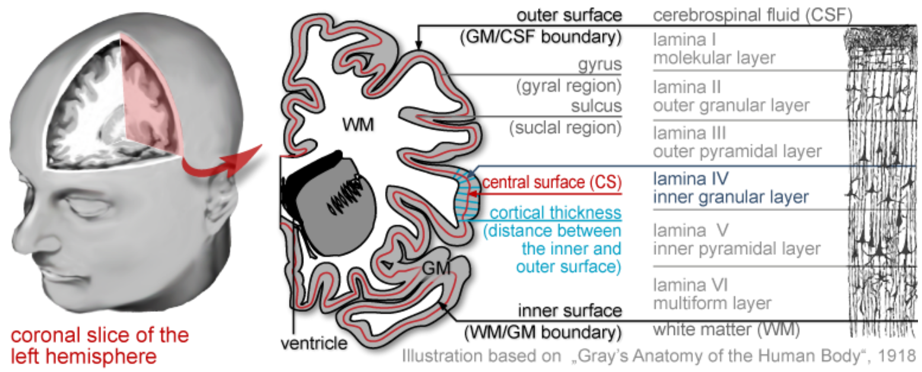


Figure 9. CAT12's CT estimation (in blue) using PBT to locate the central surface (in red) [40]

The first step of the structural preprocessing pipeline was the conversion of the multiple DICOM files belonging to each subject to a single NIfTI file, to standardize the data format. The segmentation module was then performed, which included brain tissue segmentation, denoising, intensity adjustments, skull-stripping, cleanup (removal of remaining meninges), spatial normalization (where images are aligned to the standard MNI space), CT estimation, and volume ROI processing (see Figure 9). The latter was computed under the 2018 Schaefer atlas to obtain a measure in ml for white and grey matter volume in 200 regions, although white matter will not be studied in this project.

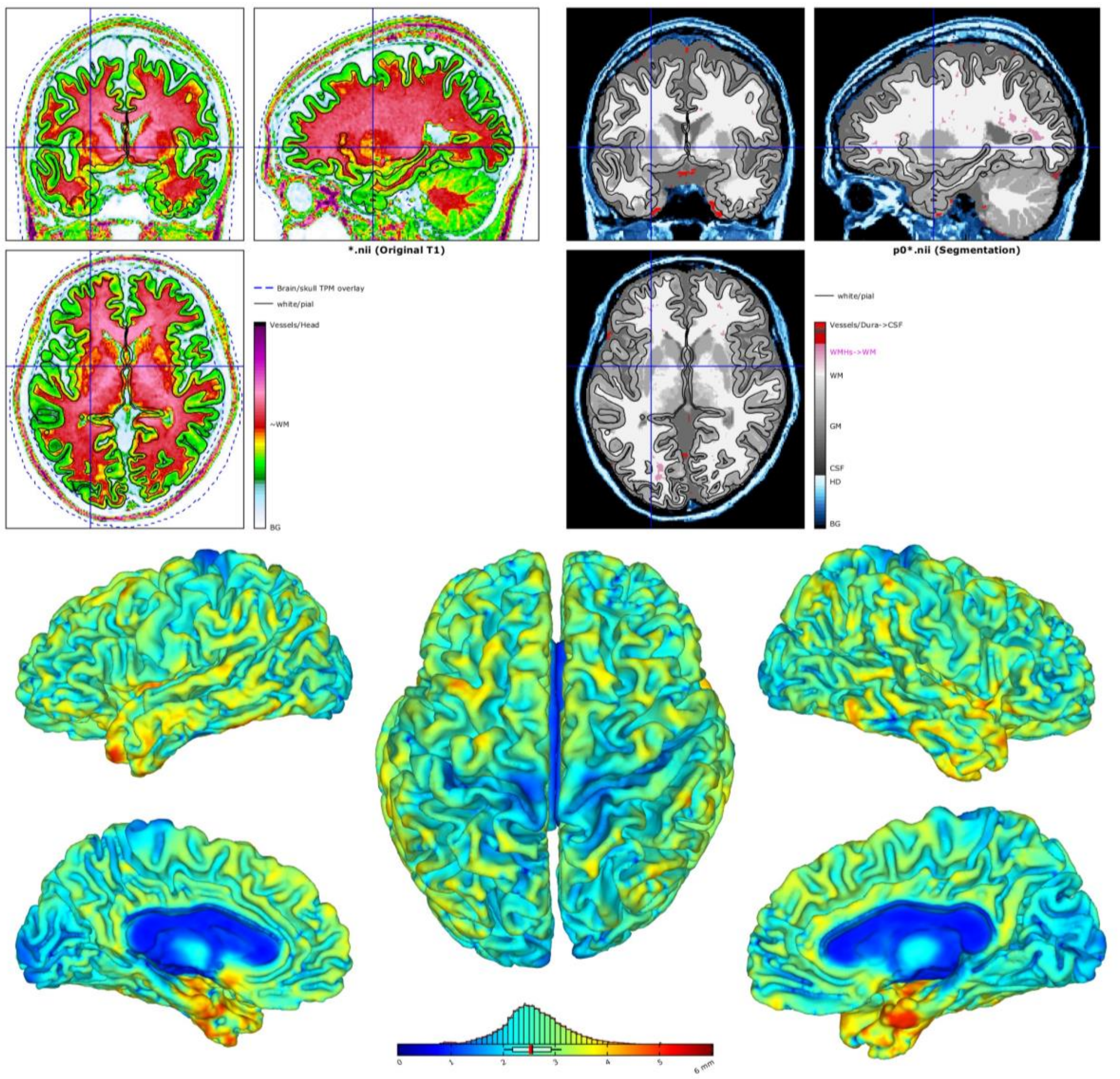


Figure 10. Results of brain tissue segmentation (top right) and CT estimation (down) for subject 005_S_6093 of the CN subject group

Despite being less commonly recognised within the neuroscience literature than other atlases such as the Desikan-Killiany or the AAL, the 2018 200 Parcels Schaefer offers a very suitable balance between anatomical and functional relevance for this study. Also known as Local-Global Intrinsic Functional Connectivity Parcellation, this functional atlas includes parcellations at multiple resolutions (100, 200, 300, 400, 1000 parcels) based on functional connectivity obtained from both task-fMRI and rs-fMRI data of 1489 participants (see Figure 11) [41]. Developed through a gradient-weighted Markov Random Field (gwMRF) model, the resulting parcellations showed agreement with the boundaries of certain cortical areas defined using histology and visuotopic fMRI, indicating fine compatibility with structural measures. Additionally, it was recognised that this atlas can potentially be useful for future applications with computational limitations requiring dimensionality reduction of voxel-wise fMRI.

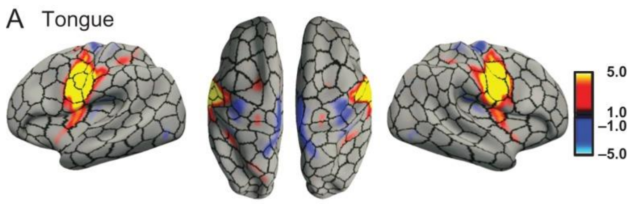


Figure 11. Group-average task activation map of tongue motion using a 400 parcels resolution [41]

The segmentation module also includes a feature that automatically generates quality reports (see Figure 12 as an example). These reports were used to discard images with less than 80% resolution, and can be found under the “Annex 1” folder in the GitHub repository.

CAT-Segmentation: ..tal_IR-FSPGR\2017-10-27_10_55_34.0\I924228\s005_S_6093-0004-00001-000001-01.nii

Version: OS / Matlab / SPM12 / CAT12 / seg:	WIN / 23.2 / 7771 / 12.9 (2560) / 1639
Tissue Probability Map:	\color[rgb]{0 0 0}..sama\Documents\MATLAB\spm12\tpm\TPM.nii
Optimized Shooting Registration to:	..s_MNI152NLin2009cAsym\Template_0_GS.nii
affreg / APP / setCOM	mni / default / COM
biasstr	medium
LAS strength / Skull-Stripping:	medium / APRG
WMH Correction / Int. Res.:	(WMH=WM) / optimal(1.00 0.30)
Voxel resolution (original > internal > PBT; vox):	1.00x1.00x1.00 > 1.00x1.00x1.00 > 0.50^3 mm^3; 1.50^3 mm^3

Image and Preprocessing Quality:	
Resolution:	**85.00% (B)**
Noise:	**93.54% (A)**
Bias:	**95.15% (A)**
Weighted average (IQR):	**86.63% (B)**
Mean surface Euler number:	20
Defect area:	0.40%
Processing time:	114:12 min

Volumes:	CSF	GM	WM
Absolute volume:	322	650	472 cm^3
Relative volume:	22.3	45.0	32.7 %
TIV:	1444 cm^3		
Thickness:	2.61±0.63 mm		

Figure 12. Quality check parameters (resolution, noise, bias) for subject 005_S_6093

Then, to extract the mean cortical thickness feature inside the 200 Schaefer parcels for regional CT estimation, the RBM module was computed using the left hemisphere thickness data that was generated in the segmentation step. The resulting ROI-based surface values were saved in mm in a CSV file, where two extra parcels that are generated for surface analysis representing the brain’s median wall were discarded (see Table 1). The mean GMV inside each ROI was also estimated using RBM to fit this format (Table 2).

Table 1. Regional cortical thickness (CT) of the first 40 brain regions (left and right hemispheres) for subject 005_S_6093

ROI_Schaefer2018_200Parcels_17Networks	Regional CT (mm)
l17Networks_LH_VisCent_ExStr_1	24.405
r17Networks_RH_VisCent_ExStr_1	26.054
l17Networks_LH_VisCent_ExStr_2	20.843
r17Networks_RH_VisCent_ExStr_2	25.341
l17Networks_LH_VisCent_Striate_1	18.555
r17Networks_RH_VisCent_Striate_1	19.034
l17Networks_LH_VisCent_ExStr_3	21.895
r17Networks_RH_VisCent_ExStr_3	23.273
l17Networks_LH_VisCent_ExStr_4	23.444
r17Networks_RH_VisCent_ExStr_4	24.188
l17Networks_LH_VisCent_ExStr_5	20.536
r17Networks_RH_VisCent_ExStr_5	21.288
l17Networks_LH_VisPeri_ExStrInf_1	27.528
r17Networks_RH_VisPeri_ExStrInf_1	21.818
l17Networks_LH_VisPeri_ExStrInf_2	23.839
r17Networks_RH_VisPeri_ExStrInf_2	20.986
l17Networks_LH_VisPeri_ExStrInf_3	22.282
r17Networks_RH_VisPeri_StriCal_1	21.072
l17Networks_LH_VisPeri_StriCal_1	22.143
r17Networks_RH_VisPeri_ExStrSup_1	21.225
l17Networks_LH_VisPeri_ExStrSup_1	22.650
r17Networks_RH_VisPeri_ExStrSup_2	22.436
l17Networks_LH_VisPeri_ExStrSup_2	22.050
r17Networks_RH_VisPeri_ExStrSup_3	22.388
l17Networks_LH_SomMotA_1	25.294
r17Networks_RH_SomMotA_1	24.040
l17Networks_LH_SomMotA_2	24.723
r17Networks_RH_SomMotA_2	22.286
l17Networks_LH_SomMotA_3	18.794
r17Networks_RH_SomMotA_3	26.973
l17Networks_LH_SomMotA_4	23.770
r17Networks_RH_SomMotA_4	19.460
l17Networks_LH_SomMotA_5	21.078
r17Networks_RH_SomMotA_5	24.384
l17Networks_LH_SomMotA_6	25.733
r17Networks_RH_SomMotA_6	22.078
l17Networks_LH_SomMotA_7	21.709
r17Networks_RH_SomMotA_7	21.532
l17Networks_LH_SomMotA_8	17.497
r17Networks_RH_SomMotA_8	24.386

Table 2. Grey matter volume (GMV) of the first 40 brain regions (left hemisphere, abbreviated) for subject 005_S_6093

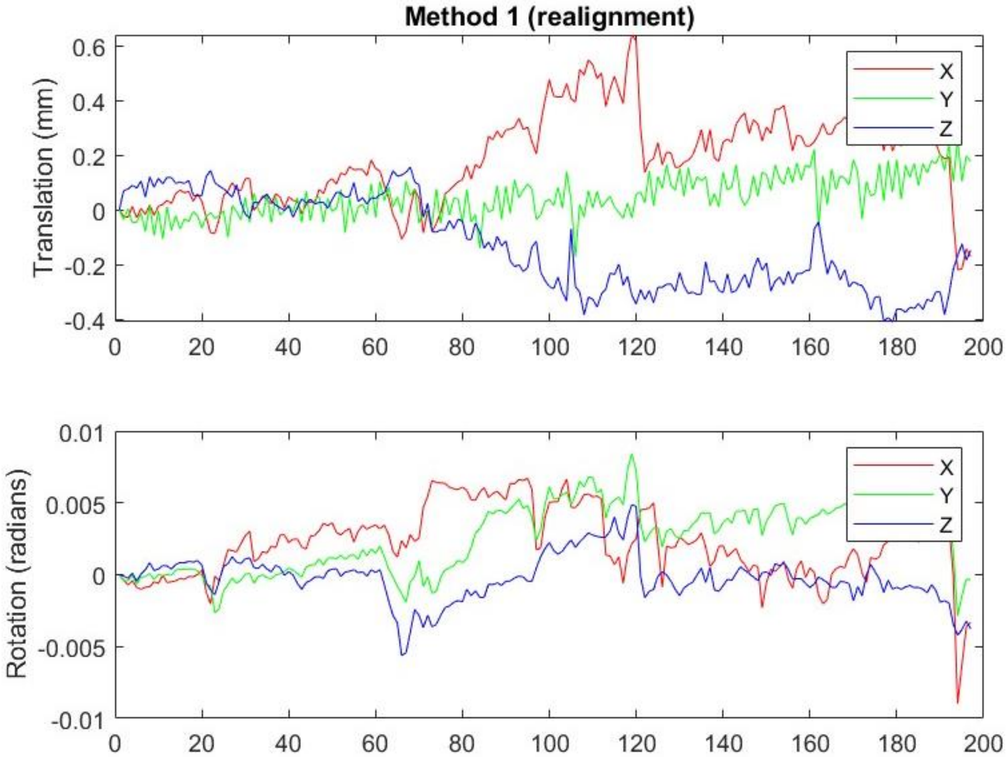
ROI_Schaefer2018_200Parcels_17Networks	GMV (ml)
lVisCent_ExStr_1	37.140
lVisCent_ExStr_2	19.272
lVisCent_Striate_1	26.753
lVisCent_ExStr_3	20.433
lVisCent_ExStr_4	27.780
lVisCent_ExStr_5	26.983
lVisPeri_ExStrInf_1	23.449
lVisPeri_ExStrInf_2	19.394
lVisPeri_ExStrInf_3	0.5629
lVisPeri_StriCal_1	33.127
lVisPeri_ExStrSup_1	0.8629
lVisPeri_ExStrSup_2	36.257
lSomMotA_1	17.753
lSomMotA_2	12.628
lSomMotA_3	15.914
lSomMotA_4	19.899
lSomMotA_5	10.341
lSomMotA_6	18.202
lSomMotA_7	29.917
lSomMotA_8	0.7110
lSomMotB_Aud_1	24.092
lSomMotB_Aud_2	27.295
lSomMotB_S2_1	19.622
lSomMotB_S2_2	26.691
lSomMotB_Aud_3	17.782
lSomMotB_S2_3	16.651
lSomMotB_Cent_1	25.980
lSomMotB_Cent_2	17.965
lDorsAttnA_TempOcc_1	47.321
lDorsAttnA_TempOcc_2	21.022
lDorsAttnA_ParOcc_1	22.333
lDorsAttnA_SPL_1	18.905
lDorsAttnA_SPL_2	21.403
lDorsAttnA_SPL_3	13.909
lDorsAttnB_PostC_1	14.202
lDorsAttnB_PostC_2	0.9078
lDorsAttnB_PostC_3	16.088
lDorsAttnB_PostC_4	22.900
lDorsAttnB_FEF_1	22.258
lSalVentAttnA_ParOper_1	25.796

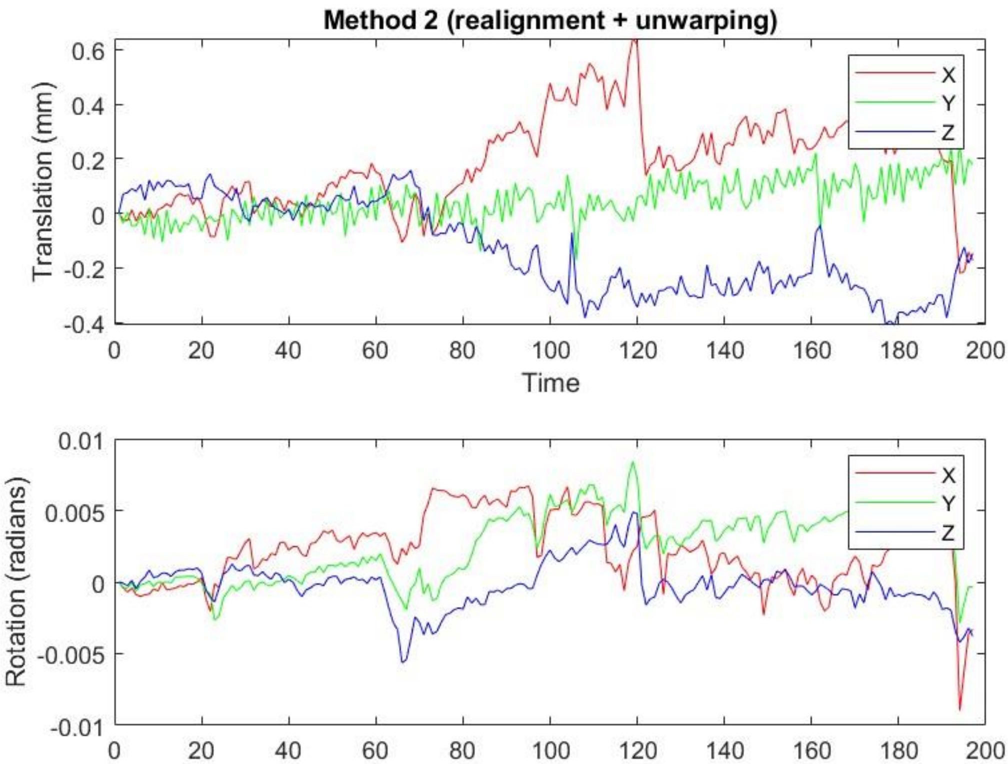
2.2.2 Resting-State Functional MRI Preprocessing

Functional connectivity (FC) is defined as the temporal similarity between spatially remote neurophysiological events, providing insights into both the specialization and interaction of brain networks [42]. When studied under methods like seed-based analysis, independent component analysis (ICA), or even graph analysis, this second measure (often referred to as "functional integration") can be obtained. The most common way to quantify FC for functional integration is by calculating the Pearson correlation coefficient between the rs-fMRI time series of the different ROIs [43]. However, a very extensive preprocessing accounting for the inherent artifacts and noise of these BOLD signals is crucial to ensure a reliable assessment of FC.

One of the first requirements for this preprocessing is to correct for subject motion through the realignment of each fMRI volume with a reference volume, so that each voxel represents the same brain region across all scans. To compute this, SPM12 uses a six-parameter (three for translation and three for rotation) rigid body transformation, which involves moving and rotating a voxel in space without changing its shape or size [38]. Given the 3.4 mm slice thickness of the data, a sampling distance and Gaussian smoothing kernel of 4 mm were chosen to accurately realign the images to the mean. After the rigid body transformation, a trilinear interpolation was employed to estimate the new voxel values in the transformed images, and these were then resliced onto a new coordinate space through a higher precision 2nd degree B-spline.

An additional procedure known as unwarping can also be examined to address motion-induced variance and inhomogeneities of the magnetic field, such as those caused by air pockets in the sinuses [44]. It applies a non-linear correction based on the field maps and the estimated motion parameters from realignment to finally adjust the voxel spatial positions. However, this preprocessing step is computationally demanding, so the motion corrections with and without unwarping were compared in a custom MATLAB script for ten of the samples using the generated estimate files. The plotted values turned out very similar over time for both preprocessing methods, with small fluctuations around zero, suggesting that realignment without unwarping can be effective enough for correcting motion and field artifacts (see example Figures 13 and 14). Unwarping was therefore no longer considered for this project.





Figures 13 and 14. Motion parameters after realignment (above) and realignment & unwarping (below) for subject 002_S_1261

Within the temporal preprocessing requisites, slice-timing correction is intended to account for the non-simultaneous slice acquisition that is used during echo-planar imaging (EPI) [38]. During an fMRI scan, the entire brain is imaged slice by slice in quick succession, so that a single volume is not recorded at the same time. This creates a temporal discrepancy which can affect any time-sensitive analysis, making it necessary to align the acquisition times of all slices within a single volume to the acquisition time of the first slice of that same volume. To provide such a continuous signal, the SPM software performs a Fourier transform to the BOLD signals and a phase shift to the resulting sinusoids that make up the signal, which is calculated based on the known slice acquisition order (in this case, an even/odd posterior to anterior order [45]) and the timing to acquire all the slices within a single volume. This second factor is also known as acquisition time or TA, and it is calculated as the subtraction of the time it would take to acquire a single slice (TR/ total number of slices) from the total TR. The total number of slices is specific for each scan and was consulted in ADNI's MAYOADIRL_MRI_FMRI_NFQ_26May2024 study data.

A normalization step was also necessary to align images to the standard MNI space. For that, the transformation parameters ("deformation field") derived during the earlier step of sMRI normalization and a 4th Degree B-spline were utilized. The co-registration of the Schaeffer 200 Parcels atlas to the first normalized fMRI slice of each subject was also computed using a nearest neighbor interpolation. Finally, the signal underwent an additional convolution with an 8x8x8 mm smoothing filter to increase the signal-to-noise ratio in case of any additional physiological noise remaining. The Gaussian kernel size was chosen based on a common rule of thumb, which suggests using a smoothing filter approximately three times the slice thickness [46].

Once the necessary preprocessing steps had been completed, a custom script was designed based on SPM's Volume of Interest analysis [47] to obtain the FC between the 200 ROIs. For that, the BOLD signal intensity was averaged across all voxels within a region at each time point. The Pearson correlation coefficient was then calculated between each pair of time series (measured in Arbitrary Units of Intensity), and the Fisher r-to-z transformation was applied to convert each coefficient to z-score. This is a common practice in FC studies employing machine learning models, as the transformation stabilizes the variance of the correlation values and makes them more normally distributed [48, 49]. Following Xin Di et.al quality control protocol [50], the resulting connectivity matrix was plotted along with other parameters to check the relation of the time series to head motion and physiological noise: global mean signal intensity across time, pairwise variance between consecutive timepoints, and rigid body motion. Matrices with incoherent block-like structures, low symmetry around the diagonal, or very high positive correlations were further examined (see Figure 15).

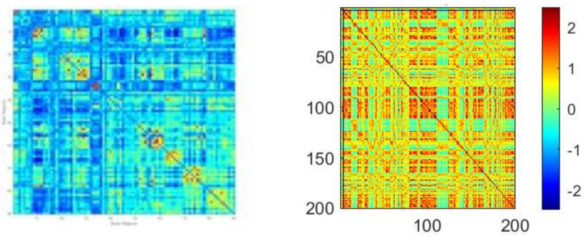


Figure 15. FC matrix for 90 ROIs found in the literature [51] (left), against the FC matrix for 200 ROIs of the discarded sample 011_S_7028 (right). Notice the high correlations outside the diagonal

For example, unusually high correlation patterns were observed for subject 003_S_4288, likely linked to a strong global mean intensity and variance (Figure 16).

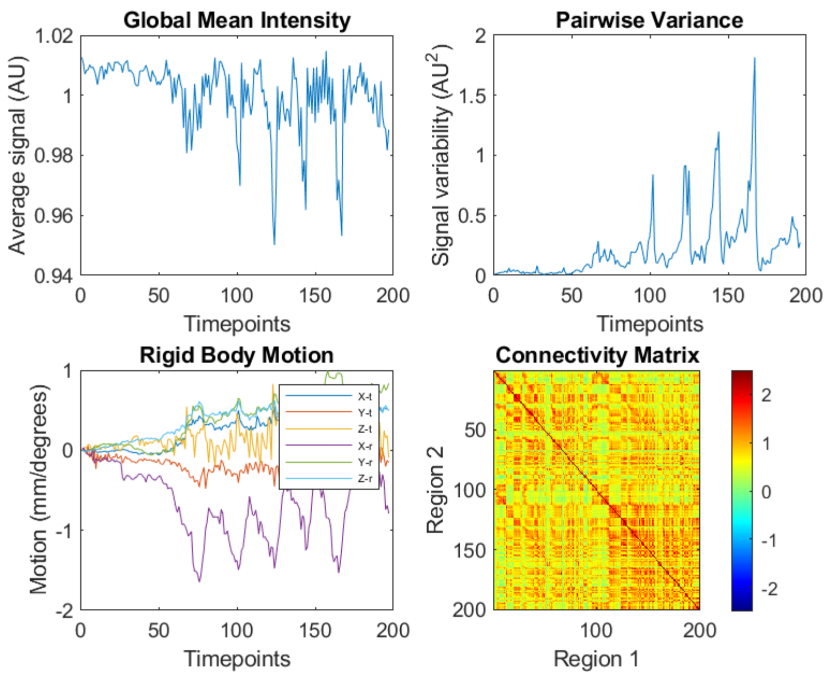


Figure 16. Global mean intensity (normalized), pairwise variance (normalized by global mean intensity), rigid body motion and connectivity matrix for subject 003_S_4288

Given that both the global mean and pairwise variance is normalized by the global mean intensity, the variance is expected to be relatively stable unless there are changes in the signal that aren't proportional to the global mean, such as scanner artifacts, periods of intense neural activity, or usually, head motion [50]. Additionally, some of these peaks (around timepoints 130 and 165) were aligned with significant changes in rigid body motion. Following these observations, scrubbing ("a term that refers to the practice of discarding incorrect or untrustworthy pieces of information" [52]) was applied to 27 timepoints (volumes) in the sample, with the hope of mitigating the effects of artifacts. Observables changes were detected in the correlation matrix after this transformation, with a decrease of high positively correlated values (see Figure 17).

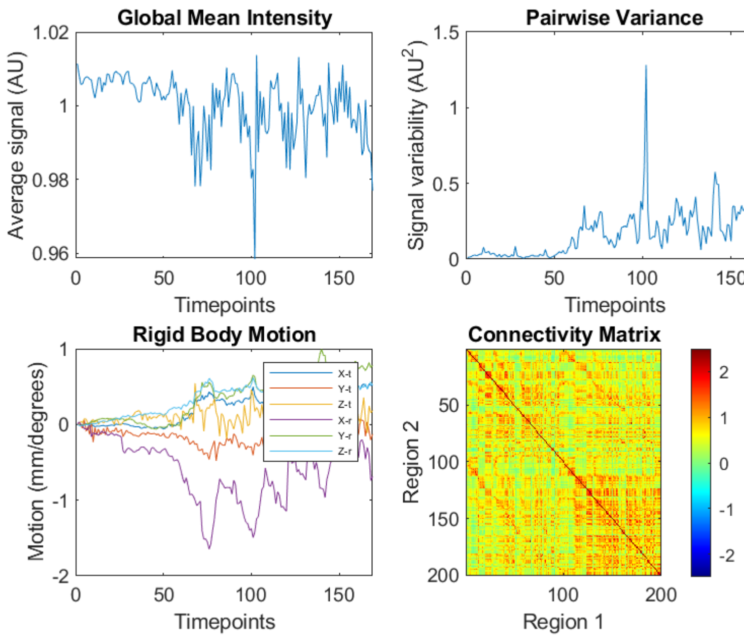


Figure 17. Global mean intensity (normalized), pairwise variance (normalized by global mean intensity), rigid body motion and connectivity matrix for subject 003_S_4288 after scrubbing

This same scrubbing process was performed for other 32 samples, always leaving at least 5 minutes of total fMRI data for reliability (given our 3s TR, a minimum of 100 volumes) [53]. From these samples, 24 of them improved significantly, while the other slightly decreased their quality.

On the other hand, samples with odd correlation values displaying rigid body motion above 3 mm/ 2 degrees or no apparent abnormal values in the rest of parameters were again preprocessed using unwarping, according to the traditionally acceptable 1.5

mm limit of root mean square (RMS) subject motion [50]. Although no changes in the rigid body motion plot could be observed after, correlation values showed a general increase of quality (see Figures 18 and 19).

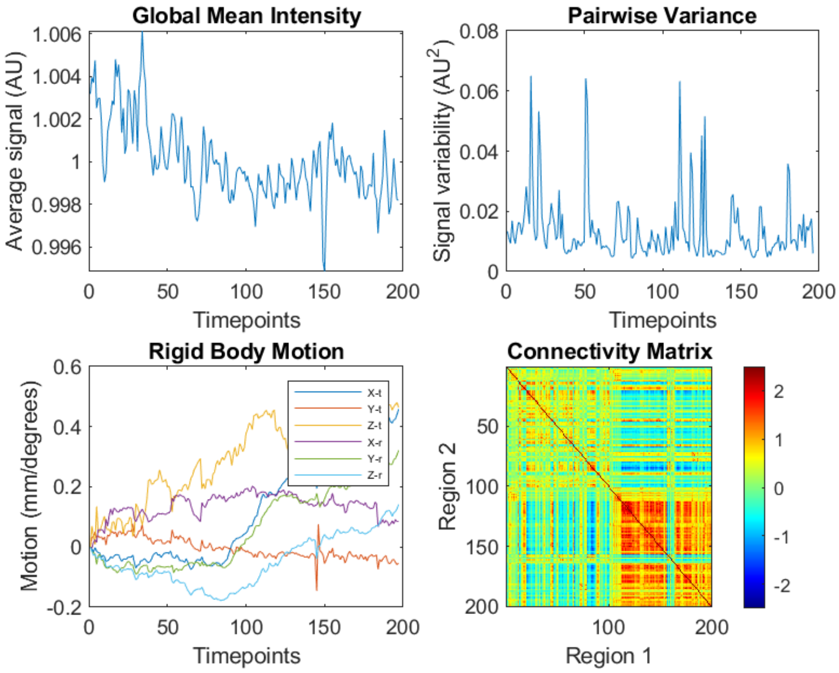


Figure 18. Quality parameters for subject 003_S_6067 before unwarping

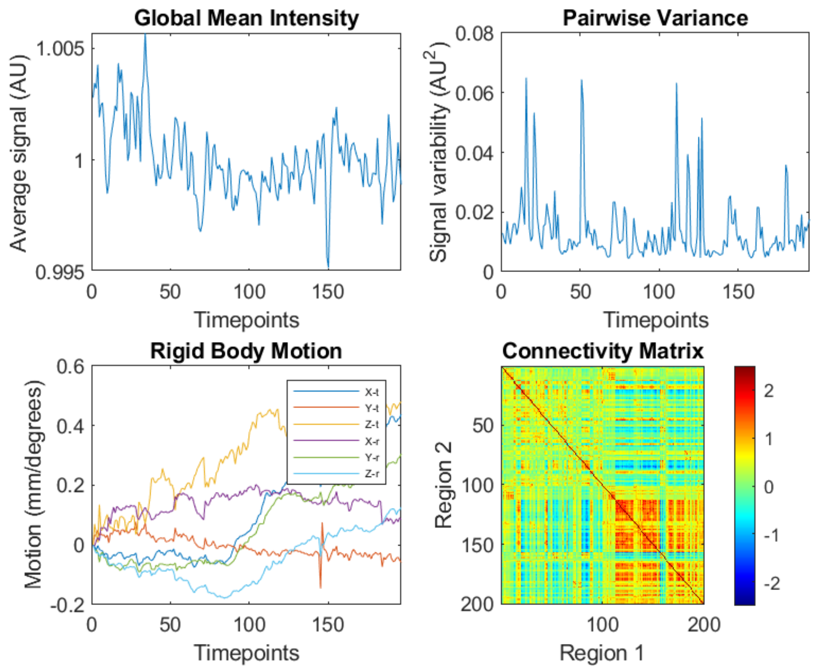


Figure 19. Quality parameters for subject 003_S_6067 after unwarping

Unwarping was also performed for four more samples, showing a slight to no improvement. As there were no significative changes for these other cases, unwarping was once again left out of the preprocessing.

Consequently, in order to correct high rigid body motion without relying to unwarping, scrubbing was also utilized for three of the samples at specific timepoints. For instance, around 80 volumes of data with more than 3 mm of subject motion in the X-rotation axis were discarded for subject 020_S_6513 (see Figure 20).

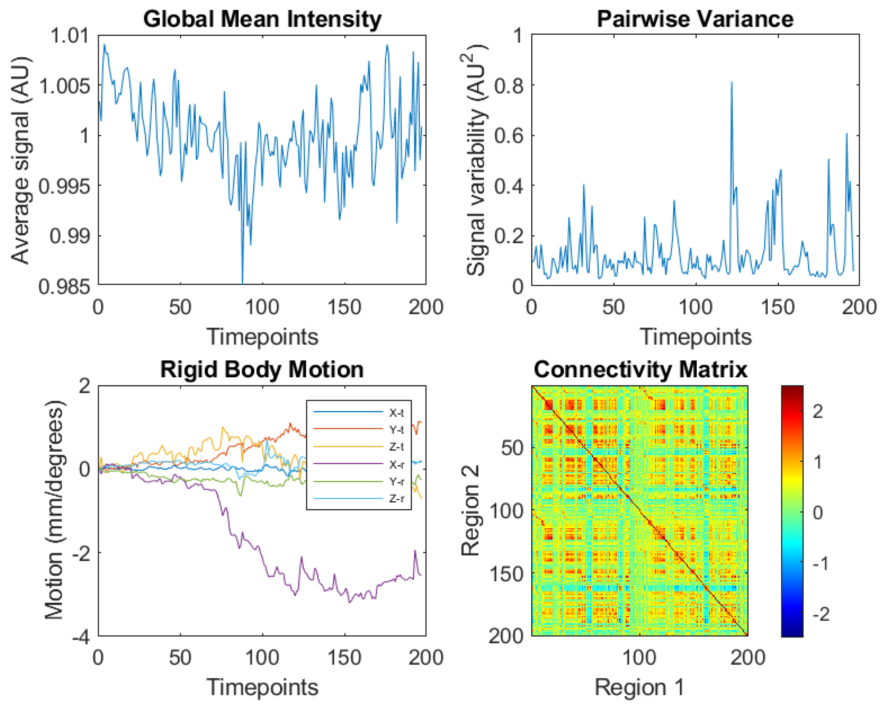


Figure 20. Quality parameters for subject 020_S_6513 before scrubbing

Unlike expected, the quality of these samples significantly worsened as it introduced changes in the signal variance (see Figure 21), and all volumes prior to scrubbing were therefore conserved.

The results from using scrubbing and unwarping to correct high rigid body motion suggest that subject motion in our data did not have a significant effect on the integrity of the FC measurements.

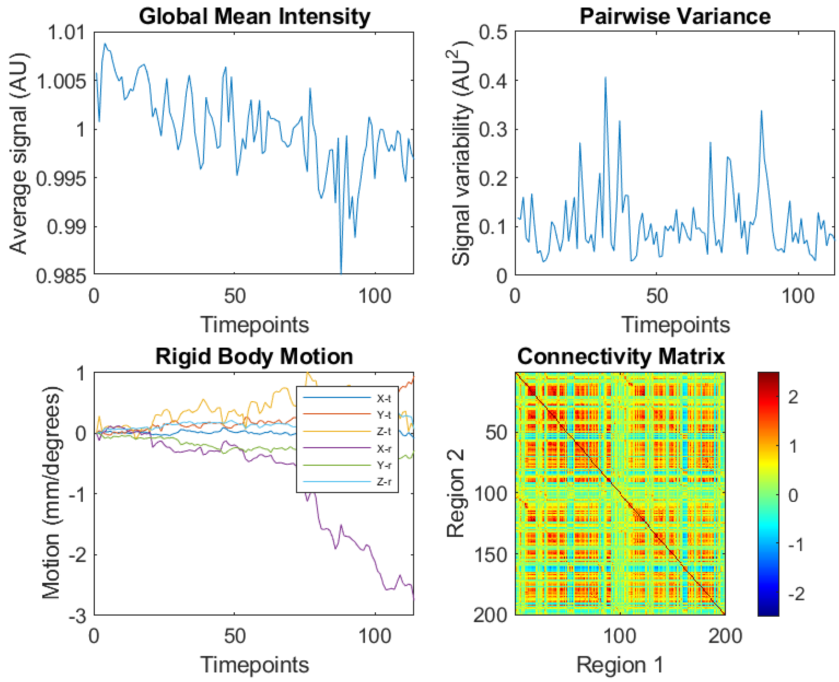


Figure 21. Quality parameters for subject 020_S_6513 after scrubbing

A third quality check scenario was also considered in several cases where blocks in the FC matrix were not very defined and correlation seemed abnormally high, possibly related to a loss of spatial resolution due to over-smoothing. However, inconclusive results were obtained when omitting the last smoothing step of the preprocessing steps (see Table 3). Moreover, it was found that adjusting the FWHM (Full Width at Half Maximum) kernel to lower values than 8 mm gave arbitrary results. The goal was to decrease the width of the signal spectrum at the points where the amplitude is half its maximum, so neighboring voxels would take a higher weight in the calculation of each voxel's intensity [54].

Table 3. Quality check for non-smoothed samples compared to their smoothed version: "I" ("improvement"), "NI" ("no improvement"), "W" ("worsening"), "E" ("excluded"), "NE" ("not excluded")

Sample subject	Observation	Sample exclusion
003_S_6092	NI	E
003_S_6257	NI	NE
003_S_6260	NI	E
003_S_6996	NI	NE

003_S_7010	NI	NE
003_S_6053	W	NE (kept the smoothed one)
003_S_0908	I	NE
003_S_6258	I	NE
003_S_6678	W	NE (kept the smoothed one)
003_S_6924	I	NE
003_S_6996	I	NE
003_S_7010	I	NE
011_S_4105	W	NE (kept the original one)
011_S_4278	W	NE (kept the original one)
011_S_6418	I	NE
014_S_4401	I	NE
014_S_6148	I	NE
020_S_6449	I	NE

As the last case scenario, signals displaying higher-than-usual correlation values, with and without smoothing, and a stable global mean intensity were directly excluded from the study (Figure 22).

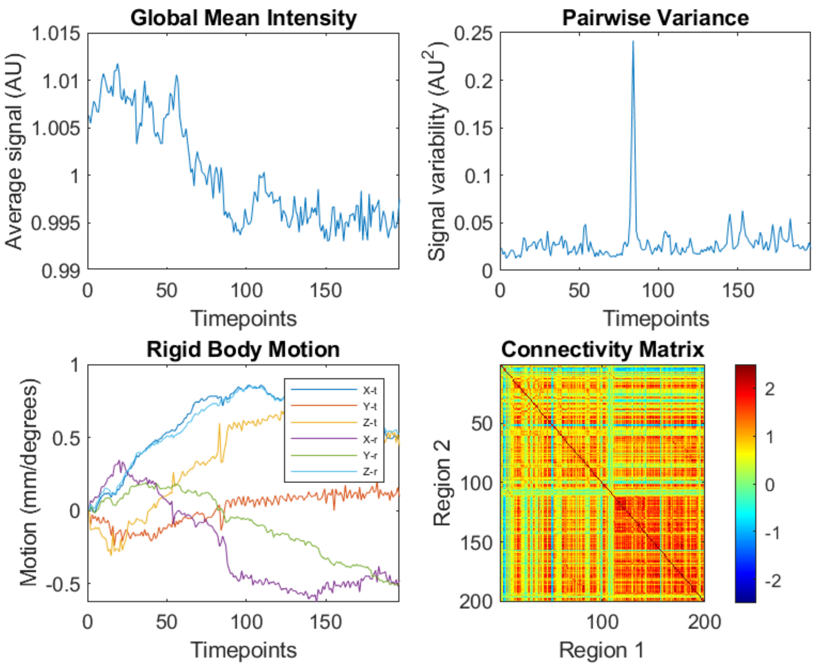


Figure 22. Quality parameters for the discarded subject 020_S_6513

All quality check reports, as well as the final FC matrices for each subject (see Table 4 as an example), can be found under "Annex 2" and "Annex 3" of this project's repository, respectively.

Table 4. FC correlation values between the first ROI and the 10 first ROIs, for the CN subject 002_S_6030

Region 1	Region 2	Fisher Correlation
1	1	Inf
1	2	0.878
1	3	1.657
1	4	0.460
1	5	0.651
1	6	0.739
1	7	0.746
1	8	1.452
1	9	0.726
1	10	1.237

Once quality was ensured, the next step was to obtain a single FC feature between regions, so the median value of the transformed correlations was calculated across all subjects in each subject group (see Tables 5 and 6). The resulting tables were saved in two separate CSV files under "Annex 3", as well.

Table 5. Median FC correlation values between the first ROI and the 10 first ROIs, for CN subjects

Region 1	Region 2	Fisher Correlation
1	1	Inf
1	2	0.745
1	3	1.119
1	4	0.778
1	5	0.925
1	6	0.899
1	7	0.856
1	8	0.995

1	9	0.530
1	10	0.900

Table 6. Median FC correlation values between the first ROI and the 10 first ROIs, for MCI subjects

Region 1	Region 2	Fisher Correlation
1	1	Inf
1	2	0.850
1	3	1.167
1	4	0.854
1	5	0.963
1	6	0.895
1	7	0.911
1	8	1.011
1	9	0.586
1	10	0.901

2.3 GNN WORKFLOW

This chapter details the workflow behind the proposed GNN model for MCI/ CN identification, separating it into graph construction and modelling (see Figure 23).

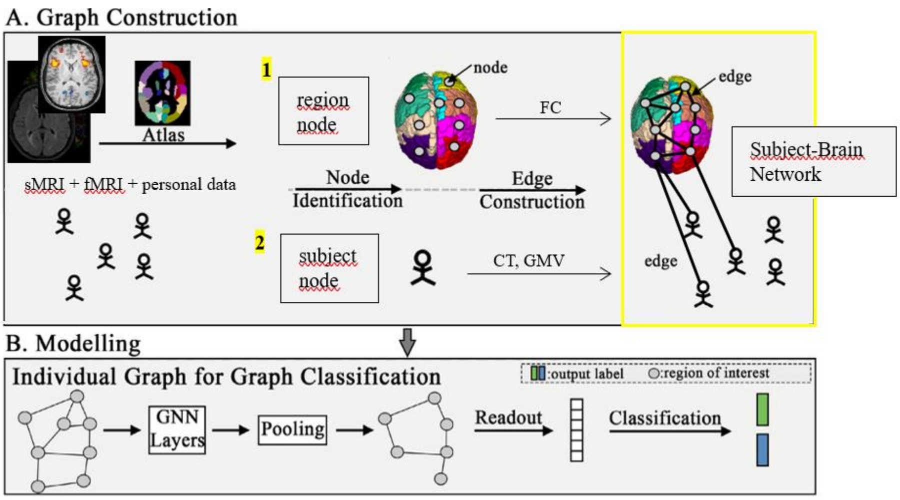


Figure 23. GNN workflow schema (inspired by L. Zhang et. al [24])

2.3.1 Graph Construction

Once feature extraction had been completed, a graph database representing our subject-brain network was constructed using the Neo4j AuraDB Free commercial software.

AuraDB is Neo4j's fully automated cloud graph platform service that "helps build intelligent, context-driven applications faster with lightning-fast queries, real-time insights, built-in developer tools, and data visualization" [55]. Instead of following a relational model, AuraDB is built around a native graph database which stores and manages data (supporting all CRUD operations) as labelled nodes, edges which connect nodes, and attributes of nodes and edges. Integrating the ACID (Atomicity, Consistency, Isolation, and Durability) properties in its transaction processing [56], as well as a property graph approach, AuraDB is designed to additionally enhance traversal performance and operational efficiency. The service uses Cypher, one of the most widely used open graph query languages, whose intuitive syntax and similarity to SQL clauses allows for easy writing and maintenance of queries.

The design of this project's graph database consists of the identification of two labelled node types: 200 region nodes corresponding to the ROIs of this study, and 152 subject nodes. Properties of the region nodes include an id and the name of the region in the 2018 200 Parcels Schaeffer atlas, while for the subject nodes, these cover id, age, sex, and diagnosis. Two types of edges were also defined using the s-MRI and f-MRI data: one to connect subject and region nodes through regional CT and GMV attributes, and another to connect regions to other regions through their FC values for each subject group (see Figure 14).

[Subject] --(HAS_REGION: ct, gmv)--> [ROI]

[ROI] <-(FUNCTIONALLY_CONNECTED_TO: corr_mci, corr_cn)-> [ROI]

Figure 24. Graph notation for the two types of edges

One of the easiest ways to import data directly into Neo4j Browser is through web-hosted CSV files, by separating node and edge types into different tables [56]. This was performed using a table for each entity of the proposed graph from the GitHub repository: “regions”, “subjects”, “has_region”, and “is_functionally_connected”. To create the region and subject tables including all their properties, visit information of all samples from the ADNI dataset and CAT12’s 2018 Schaefer atlas map were downloaded, respectively. On the other hand, the files with CT and GMV values were converted to columns in a single CSV table representing the first edge, and so were the ones containing the FC features for the second edge. In the case of FC, the infinite values of the Fisher correlation had to be converted into a numerical value (3 and -3) for compatibility with Neo4j (all Cypher queries can be found under “Annex 4”). The resulting nodes and edges (limited to 25 elements each) and data schema provided by Neo4j’s visualization tools ([57]) can be observed below (Figures 25, 26 and 27):

Figure 25. Neo4j database schema

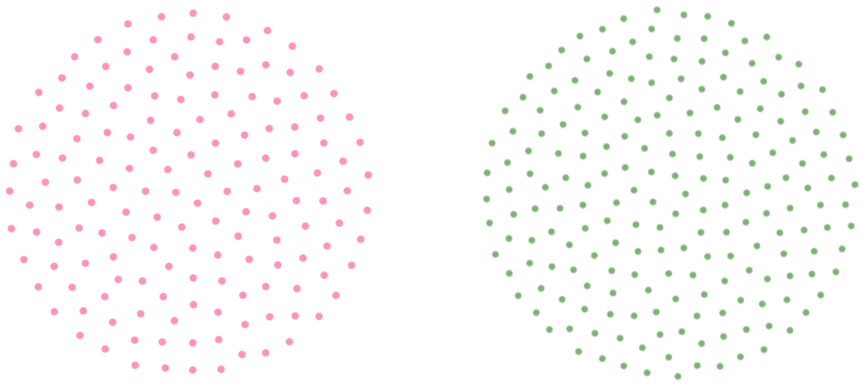


Figure 26. Subject (left) and region (right) nodes

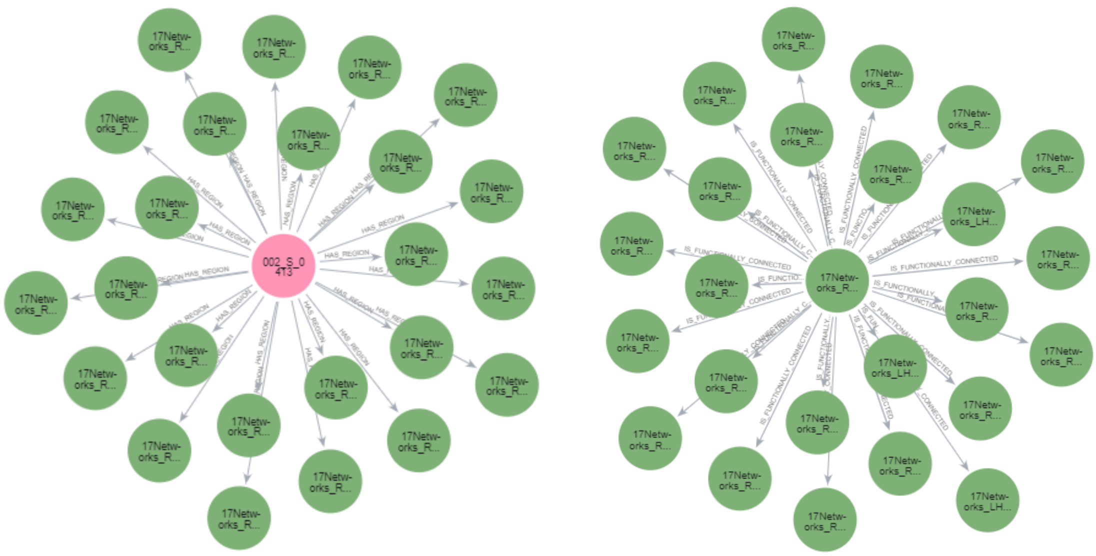


Figure 27. Subject-Region "HAS_REGION" edge (left) for subject 002_S_0413, and Region-Region "IS_FUNCTIONALLY_CONNECTED" edge (right) for region with name "17Networks_RH_DorsAttnB_PostC_4"

2.3.2 Modelling

The presented modelling problem is designed as a binary node classification, where a label $\in \{0, 1\}$ is assigned to each graph node which describes the diseased (MCI) or healthy (CN) status of the subject. Additionally, a semi-supervised strategy is adopted, where all node and edge features of the heterogeneous graph are fed to the GNN, while only a subset (the subject nodes) is labelled during training and used for the optimization process. The basis of GNNs, as well as both convolution algorithms will be explained below:

A GNN is a connectivity model that uses the information that is passed between nodes in a graph to learn and decode their pairwise relationships. For that, a graph needs to be mathematically defined as $\boldsymbol{G} = \boldsymbol{(V, E)}$, where $v \in \boldsymbol{V}$ denotes the nodes, and $e \in \boldsymbol{E}$ denotes the edges [25]. These can be expressed in terms of a $\boldsymbol{X} = \boldsymbol{N}\times\boldsymbol{D}$ feature matrix (where $\boldsymbol{N}$ denotes the number of nodes, and $\boldsymbol{D}$ is the number of node attributes) and a square adjacency matrix $\boldsymbol{A}$ with dimensions $\boldsymbol{N}\times\boldsymbol{N}$, respectively [58]. The latter one is responsible for the representation of the graph structure, where every element a_{ij} is either 1 or 0 depending on whether the two nodes x_i and x_j are connected by an edge or not (diagonal elements are always set to 1 indicating a self-connection) [59].

The goal of GNNs is to learn from this representation of data, by updating the node matrix using the adjacent neighbor's information during a procedure called message-passing, which is executed across multiple layers (***l***), At each layer, the features of a node are updated by aggregating information from its neighboring nodes and applying various transformations (see Figure 28).

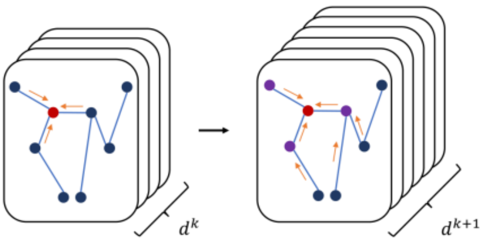


Figure 28. Schema of a GNN message passing across different k layers [25]

The mathematical formula can be described as follows, where $\boldsymbol{H^0} = \boldsymbol{X}$ (the input) and $\boldsymbol{H^L} = \boldsymbol{N}\times\boldsymbol{F}$ (the output, with $\boldsymbol{F}$ as the number of output features per node), and $\boldsymbol{f(\cdot,\cdot)}$ refers to aggregation, normalization, linear transformation and activation:

$$H^{(l+1)} = f(H^{(l)}, A)$$

Aggregation involves including information from a node's neighbor, typically through a simple sum calculation. This is computed through the multiplication of the adjacency matrix times the node matrix, producing an updated node matrix with each node vector equal to the sum of its neighbour node vectors with itself [59]. It is also very common to use the average, as it can be observed in Figure 29:

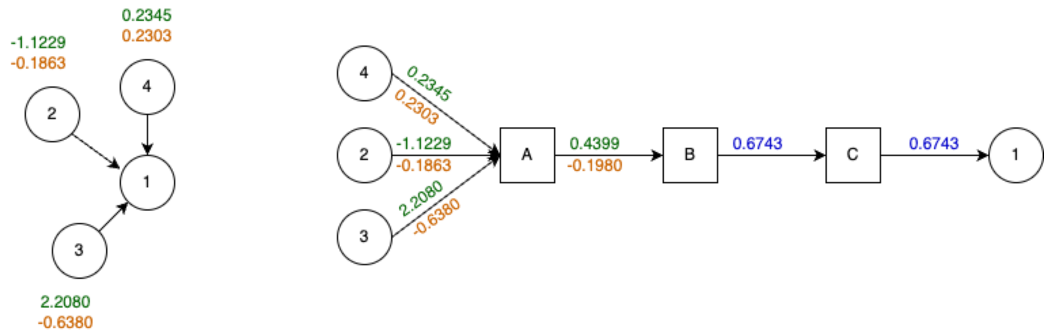


Figure 29. Schema of the aggregation function for homogeneous graphs, where information is propagated from the source nodes to target node 1 by taking the mean [60]

Here, it is also necessary to apply a normalization step to the adjacency matrix to scale the feature vectors of the nodes, ensuring that nodes with more neighbors do not introduce bias during aggregation. This is done by multiplying it with the inverse of the diagonal degree matrix, which denotes the degree (or number of neighbors):

$$H^{(l+1)} = D^{-\frac{1}{2}} A D^{-\frac{1}{2}} H^{(l)}$$

After these steps, the node features are linearly transformed by multiplying the product of the last calculation ("convolution") by a learnable weight matrix $\boldsymbol{W^l}$. Once the transformation is applied, a non-linear activation function $\boldsymbol{\sigma}$ (such as the ReLU) is used to introduce non-linearity to the model and improve its expressive power, so the final formula could be represented as follows:

$$f(H^{(l)}, A) = \sigma\left(\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right)$$

A pooling operation is also usually considered after multiple convolution layers, which aggregates node features into a condensed representation of the entire graph ("graph embedding") by transforming them into a single vector compatible with neural network inputs, also ensuring permutation-invariance (so different orders of nodes and edges are not treated differently by a neural network). This final vector is passed through more convolutional layers, where each one of them takes the output of the previous layer, a new weight matrix is again applied and all transformation operations are performed, and new predictions (classifications) are made (see Figure 30).

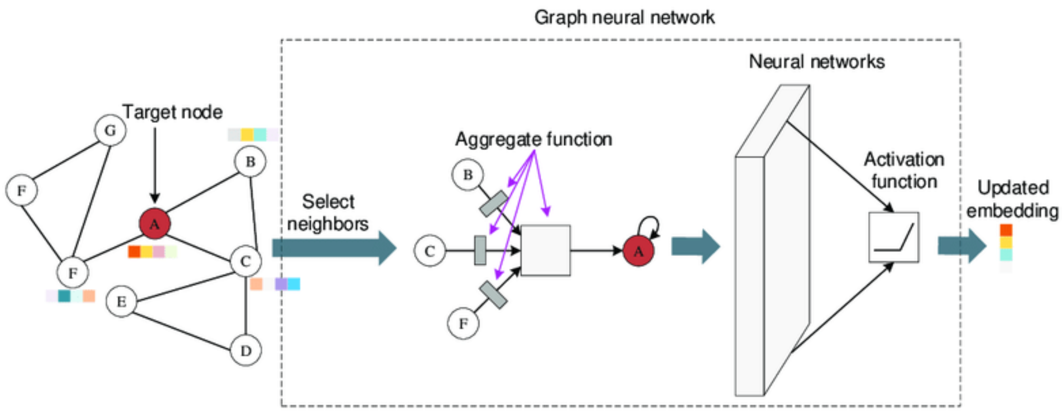


Figure 30. General schema of a GNN [61]

Having explained all of this, heterogenous graphs don't allow such a simple calculation, as the feature dimensions of one node type are incompatible with those of the other (like the subjects and regions of this project), so aggregation needs to be computed as separate operators like Graph SAGE offers, as depicted in Figure 31:

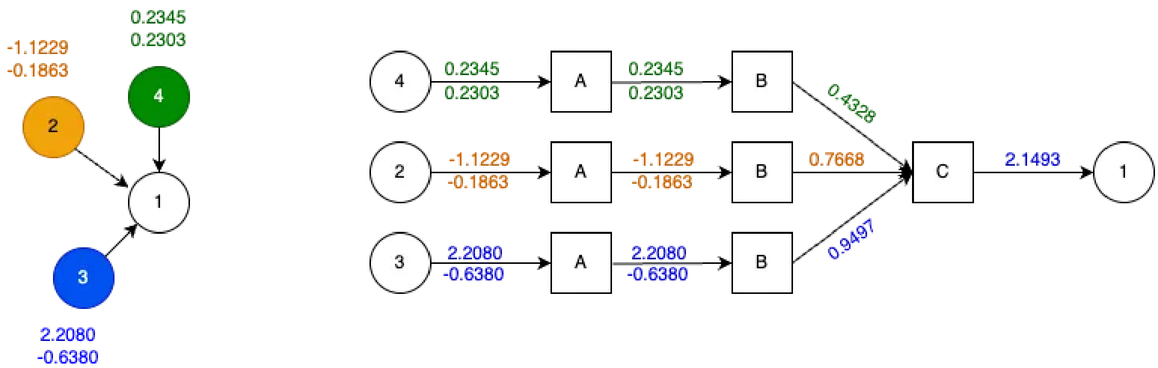


Figure 31. Schema of the aggregation function for heterogeneous graphs, where three different aggregators are utilized [62]

In some graphs, like the proposed one, edges can also have attributes, which introduces an additional level of complexity as they need to be concatenated to the node matrices.

PyTorch Geometric is an open-source library built on PyTorch that provides a framework for the creation and training of GNN-based methodologies derived from various research papers. It also supports heterogenous graph models [62] and edge attributes during message-passing in their SAGEConv function [63].

The following documentation served as basis of this model [64], together with the HeteroConv, GCNConv and SAGEConv algorithms [65, 66]. The code can be found in the GitHub repository, where details and parameters on the training, testing, evaluation and optimization steps are provided.

3 RESULTS

During the training of the proposed GNN, it was observed that the model's predictions seemed to collapse rapidly, converging quickly to zeros. This behavior indicates that the model's learning process was unstable, leading to poor performance and inaccurate predictions. This was likely caused by an inadequate message propagation, either because the edges were not well-defined or message propagation was ineffective, and the model failed to learn meaningful relationships.

Either way, an alternative representation to the Neo4j graph-based data structure was created, hoping to provide results to this project. The tabular form of data used to create the graph database was used in the model instead, and the modified code can be found here: https://github.com/IsabelMartinez2022/DetectionOfMCI_TFG_WoNeo4j Several evaluation metrics were obtained under a 5-fold cross validation from this second approach, where the model was trained on 4 of the folds and tested on the remaining fold, with each part serving as a test set in one of the iterations.

First, a summary of metrics was provided per fold in Figure 32:

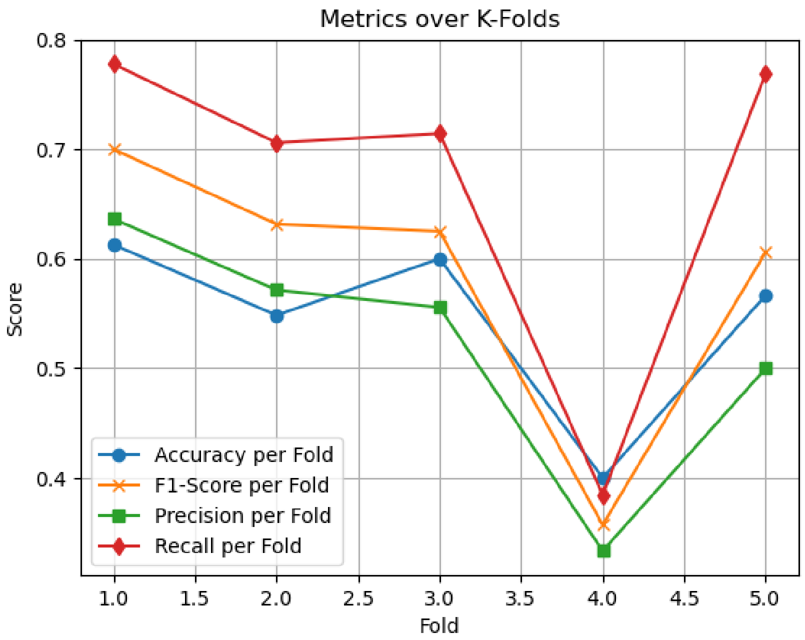


Figure 32. Accuracy (percentage of true observations out of the total), precision (true positive predictions out of all positive predictions made by the model), sensitivity ("recall") and F1-score (a metric balancing precision and sensitivity) for the 5 folds

These results evidence a very high variability and poorer performance for the fourth fold across all metrics, suggesting that the model's performance is very sensitive to specific data splits and there is a problem with the data distribution. It could be also caused by overfitting because the model might be too complex for the available data [67]. Another possible explanation behind these results could be the presence of especially noisy or abnormal data in the fourth fold.

Also, recall results are quite high, as opposed to the low precision, so the model captures most true positives but with many false positives.

In terms of costs, the following confusion matrix was obtained, where predictions in the x-axis are plotted against true values in the y-axis (see Figure 33):

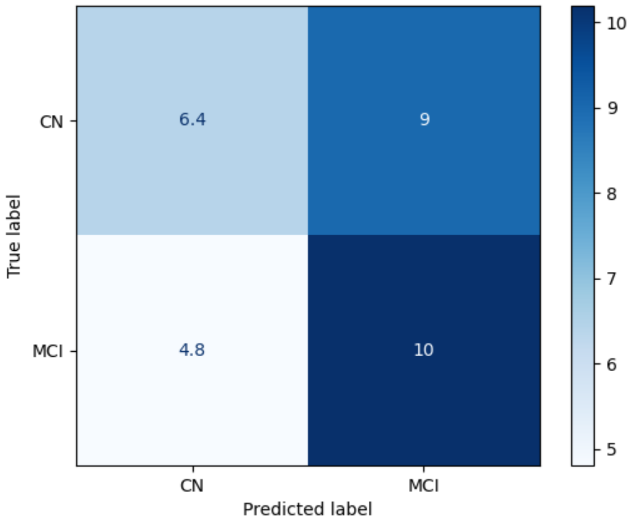


Figure 33. Average confusion matrix

The matrix indicates that the model struggles more with distinguishing CN from MCI, as it made 9 errors by classifying CN cases as MCI. Sensitivity seemed quite acceptable for MCI classification, correctly predicting an average of 10 counts across the 5 folds, but it still misclassified 4.8 MCI cases as CN. The total misclassifications (off-diagonal values: 9 + 4.8) also seems quite high compared to the correctly classified cases (6.4 + 10). When taking a closer look at each fold, it can be observed that all folds displayed a similar behaviour except for the fourth one, where CN classification performed nearly the same as MCI classification (see Figure 34).

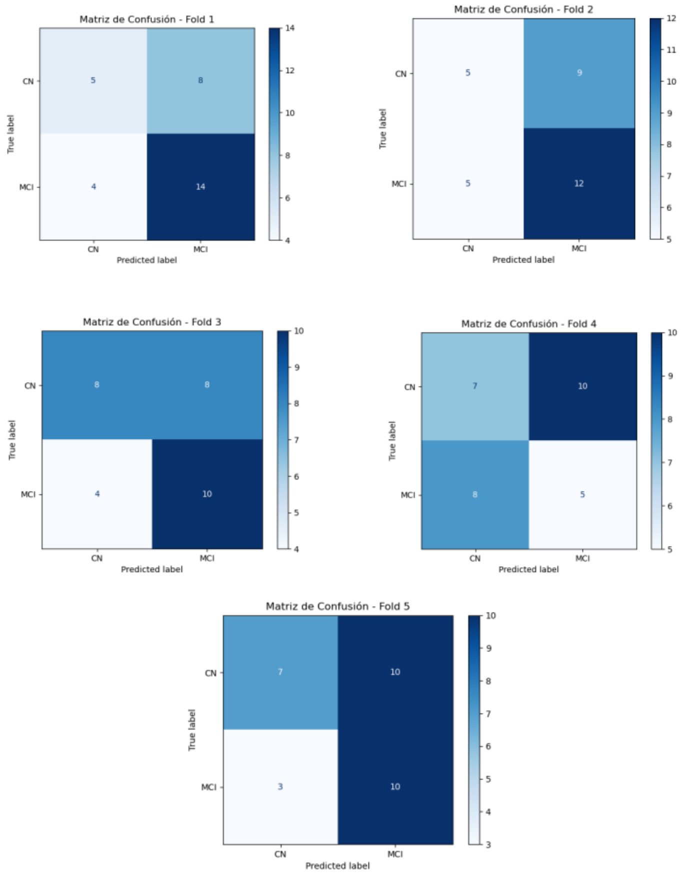


Figure 34. Confusion matrix for all 5 folds

The AUC-ROC curve provides a further insight into the costs of the model, indicating that for most folds, the model was struggling to catch significant patterns and generalize during learning, as the values are very close to 0.5 (see Figure 35).

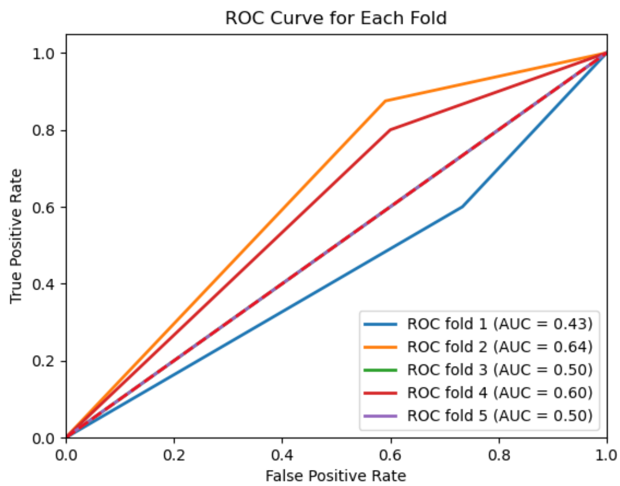


Figure 35. AUC-ROC curve, denoting measure the ability of a binary classifier to distinguish between classes [68]

4 DISCUSSION

Overall, the model exhibits a poor accuracy of 0.5456 and low precision of 0.5193, with a moderate 0.6704 of sensitivity after cross-validation.

Therefore, while these metrics indicate that the model is effective at identifying true positives, it struggles with false positives, making it very costly. Also, the ROC curves indicated AUC values ranging from 0.43 to 0.64, reflecting difficulties in consistently distinguishing between CN and MCI.

Improvements could be made trying to increase the sample size of the study to avoid overfitting, for instance, through data augmentation methods that could compensate the complexity of the GNN model. Another alternative could be applying stronger regularization methods such as the Lasso regularization (previously mentioned in this paper) that could enhance generalization.

Also, the high variability between folds could be analysed, particularly focusing on the drop of performance for the fourth fold. Noise and variability in neuroimaging data may have affected the model’s performance, so preprocessing could be furthered improved, perhaps by thresholding correlation values during the rs-fMRI preprocessing.

It should also be commented that GNNs are designed to model graph structures, and not tables. Therefore, capturing the same level of interaction through a tabular representation is not the most appropriate approach, as the effectiveness of these features in representing complex relationships can vary.

5 CONCLUSIONS

This study has explored the use of Graph Neural Networks (GNNs) for classifying subjects into CN and MCI, groups using both subject and brain connectivity and topographic data. However, a strong limitation was found during the execution of the model due to a probable wrong message propagation. This might be due to differences in how the graph data is structured or formatted in Neo4j, as opposed to Pytorch functions. An example of the integration of Neo4j in Pytorch can be found in the Neo4j's documentation [69], but further investigation on the use of more complex, heterogeneous, property graphs is required.

However, a series of improvements could be identified thanks to these project's results, such as the need for a more robust dataset and regularisation techniques, or a stricter preprocessing of the rs-fMRI. Additionally, although the results of the classification problem are not satisfactory, the model has demonstrated a relatively good ability to detect MCI cases (reflected in a recall of 0.6704) at the cost of a high false positive rate, which is hopeful.

A series of observations were also obtained from the preprocessing steps, such as the relevance of pairwise signal variation against rigid body motion, or the utility of scrubbing for improving FC correlation values.

In conclusion, this study provides a solid basis for further work improvement in classification methods within clinical neuroscience. While the use of GNNs) initially offered a promising approach to tackling MCI classification, limitations in integrating Neo4j with PyTorch were found. After switching to a tabular data format, the model's performance demonstrated a reasonable ability to detect MCI cases, but significant improvements in precision are still needed.

Future research should focus on refining the data representation and exploring more effective methods for integrating graph-based structured data.

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